ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(b)** 50 per cent	30/60.
3	**(a)** only one component sublimes	Different behaviour on heating.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
6	**(c)** iron and sulphur	Only iron is magnetic.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(c)** 18 g	50 - 32.
9	**(a)** dissolve in water, filter, evaporate	Sugar dissolves.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1).	1+1
12	Boiling point is fixed (1); Heat supplies latent heat (1).	1+1
13	Slow cooling allows ordered growth (1); Few nuclei form (1).	1+1
14	Solute may have low solubility (1); e.g. a saturated solution of chalk or CO2 in water is dilute (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1).	1+1+1
16	A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1).	1+1+1
17	Salt gives ions that neutralise the charge on particles (1); Particles aggregate (1); Soap separates (1).	1+1+1
	OR: X = 0.27 (1); Y = 0.6, Z = 0.8 (1); X is most strongly adsorbed (1).	1+1+1
18	Ink spots from document and suspects (1); Run chromatography (1); Compare bands and R_f values (1).	1+1+1
	OR: Measure boiling point 100 degrees Celsius (1); Evaporate: no residue (1); Test conductivity (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	NaCl 36 g stays dissolved at 20 degrees Celsius (solubility 36) (1); KNO3 remaining dissolved = 32 g (1); KNO3 crystallised = 110 - 32 = 78 g (1); Crystals are nearly pure KNO3 (1); Fractional crystallisation (1).	1+1+1+1+1
	OR: Particles carry charge and repel (1); Coagulation with electrolytes (1); Electrophoresis moves particles in a field (1); Dialysis separates dissolved ions (1); Examples: blood purification (1).	1+1+1+1+1
20	Spot unknown and known dyes on the same paper (1); Run in a suitable solvent (1); Compare R_f values (1); Same R_f means same dye (1); Solvent changes R_f (1).	1+1+1+1+1
	OR: Screening and sedimentation (1); Alum coagulation and filtration (1); Chlorination for bacteria (1); Desalination (distillation or reverse osmosis) for dissolved salts (1); Testing quality (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the one with R_f 0.9	Largest R_f.
2	**(d)** distillation	Boiling point.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(c)** 5 per cent (m/v)	20 g in 400 mL.
5	**(c)** dissolved impurities	Boiling point elevation.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** 500 mL	30/0.06.
8	**(d)** solvent extraction	Selective solubility.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(b)** a mixture of ethanol and water	Rising temperature.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Molecules pack into gaps (1); Interactions between molecules (1).	1+1
12	They form distinct layers (1); Miscible liquids form one layer (1).	1+1
13	12/100 x 750 (1); = 90 mL (1).	1+1
14	Suspension (1); Settles and is filtered (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add water and stir; sugar dissolves (1); Filter: sand and chalk remain (1); Evaporate for sugar; separate chalk from sand by decantation or floatation (1).	1+1+1
16	Solubility depends on the solute (1); Different adsorption on paper (1); R_f varies (1).	1+1+1
17	Pure compound moves as one band (1); Mixture components have different solubilities and attractions to paper (1); Black ink separates into several coloured bands (1).	1+1+1
	OR: Vapour contains both (1); Close boiling points (1); Fractional distillation needed (1).	1+1+1
18	Sublimation: solid to vapour; evaporation: liquid to vapour (1); Camphor vs water (1); Needs different conditions (1).	1+1+1
	OR: Crush and filter juice (1); Evaporate to syrup (1); Crystallise and centrifuge crystals (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Simple: large b.p. difference (1); Fractional: close b.p. (1); Steam: heat-sensitive substances like essential oils (1); Uses (1); Apparatus differences (1).	1+1+1+1+1
	OR: Blood: suspension/colloid (1); Centrifugation separates cells (1); Milk: emulsion colloid (1); Cream by centrifuge (1); Curdling by coagulation (1).	1+1+1+1+1
20	Petroleum gas/naphtha, petrol (1); Kerosene (1); Diesel (1); Lubricating oil, bitumen (1); Column gives a temperature gradient so fractions condense at levels (1).	1+1+1+1+1
	OR: Heat both in water bath (1); Record temperature every minute (1); Pure naphthalene stays constant at 80 degrees Celsius while melting (1); Mixture melts over a range at lower temperature (1); Plot cooling curve (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** it keeps the jacket full for efficient cooling	Complete cooling.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** increased four times	12/3.
4	**(d)** to prevent bumping and give smooth boiling	Smooth boiling.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(c)** an emulsion	e.g. milk.
7	**(b)** the second component	Ascending order.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** foam	e.g. whipped cream.
10	**(c)** boiling point at a given pressure	Fixed for pure.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heating forms iron sulphide compound (1); A magnet separates physically (1).	1+1
12	Sample dissolves in solvent in the container (1); Separation does not occur (1).	1+1
13	Vinegar particles are molecular in size (1); Soap forms micelles, larger aggregates (1).	1+1
14	Alum coagulates the suspended particles (1); They settle down faster (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sugar: crystallisation of concentrated juice (1); Oxygen: fractional distillation of liquid air (1); Boiling and solubility differences (1).	1+1+1
16	Alcohol with lower b.p. distils first (1); Composition of the remaining liquid changes (1); Final temperature approaches 100 (1).	1+1+1
17	Curve shows solubility vs temperature (1); Difference at two temperatures gives crystals (1); Choose solvents with steep curves (1).	1+1+1
	OR: Alum coagulates particles (1); They settle in clumps (1); Clear water decanted (1).	1+1+1
18	Atoms of carbon disturb the regular arrangement (1); Layers cannot slide easily (1); Hence harder (1).	1+1+1
	OR: Powdered solid dissolves faster (1); Higher temperature increases rate (1); Change only one factor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sol: solid in liquid (1); Gel: liquid in solid (1); Emulsion: liquid in liquid (1); Foam: gas in liquid (1); Aerosol: solid/liquid in gas (1).	1+1+1+1+1
	OR: Magnet first: iron (1); Sublimation: ammonium chloride (1); Residue is salt (1); Order based on properties that do not alter others (1); Check purity (1).	1+1+1+1+1
20	Add salt until some stays undissolved (1); Stir, keep at the temperature (1); Weigh an evaporating dish (1); Evaporate a known mass of solution (1); Solubility = mass of salt/mass of water x 100 (1).	1+1+1+1+1
	OR: Medicine: purification of drugs by crystallisation (1); Food: sugar, salt, milk products (1); Mining: magnetic separation, froth floatation (1); Environment: waste water treatment (1); Oil spill clean-up (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.